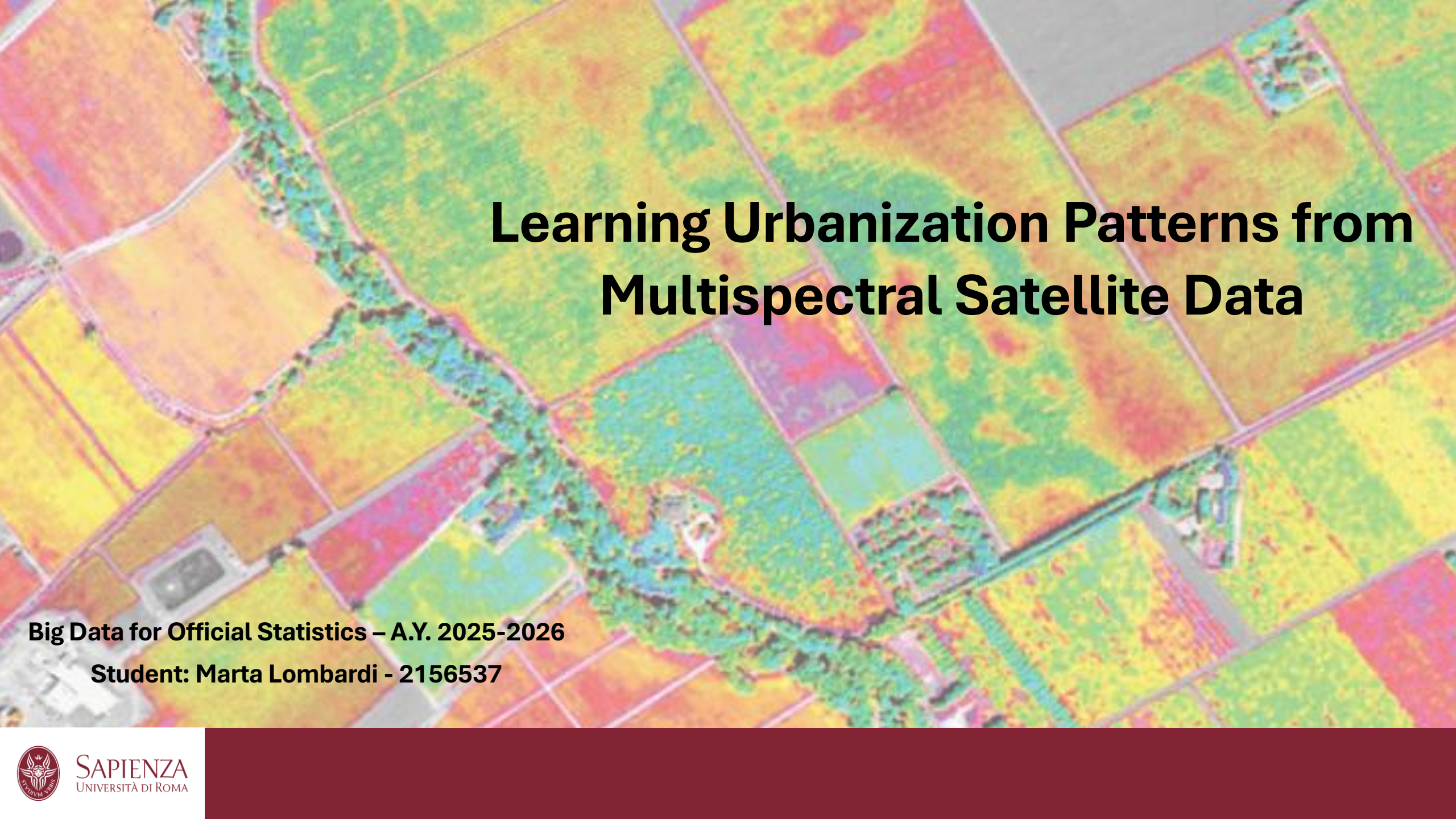




Learning Urbanization Patterns from Multispectral Satellite Data

Big Data for Official Statistics – A.Y. 2025-2026

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UNIVERSITÀ DI ROMA

From local image patches to large-scale urbanization

Objective: Learn a representation from **local satellite image patches** that captures **large-scale urbanization intensity**

Motivation

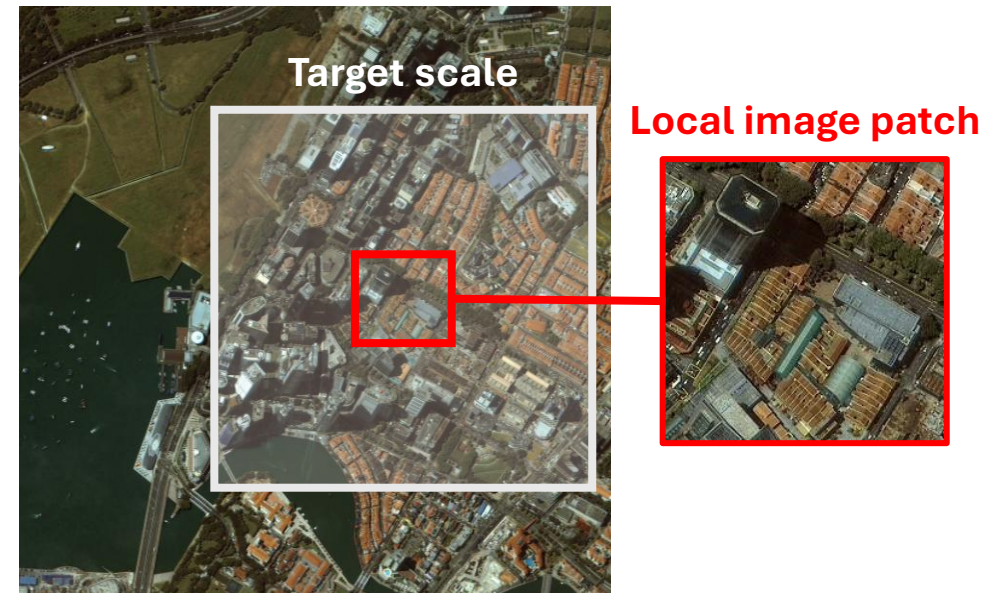
- Urbanization is a **large-scale, continuous phenomenon**.
- Available supervision (GHSL) is **coarse** and **aggregated**, while satellite images are **local** and **fine-grained**.
- Standard supervised learning struggles to reconcile these two scales.

Problem setting

- Each patch captures only **partial and localized visual evidence**
- A single urbanization value aggregates **heterogeneous land-use patterns**

Key challenges

- Similar urbanization levels may exhibit very **different local appearances**
- Similar-looking patches may correspond to **different urbanization intensities**
- **Scale mismatch**



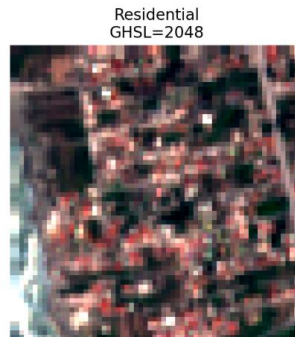
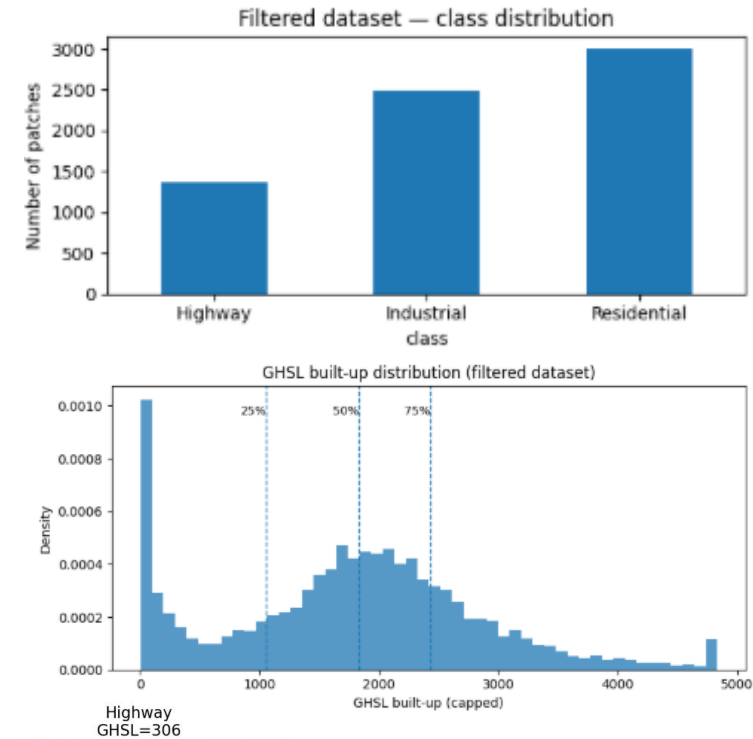
Dataset: EuroSAT and GHSL aggregation

EuroSAT (Sentinel-2 imagery)

- Collection of **multi-spectral georeferenced satellite image patches** derived from the **Sentinel-2 mission (ESA)**
- **Patch size: 64×64 pixels**
- Original dataset includes 10 land-use classes
- In this study, we focus on classes **directly informative of urbanization**, that are **Highway**, **Industrial** and **Residential**
- **Total samples used: 6857**

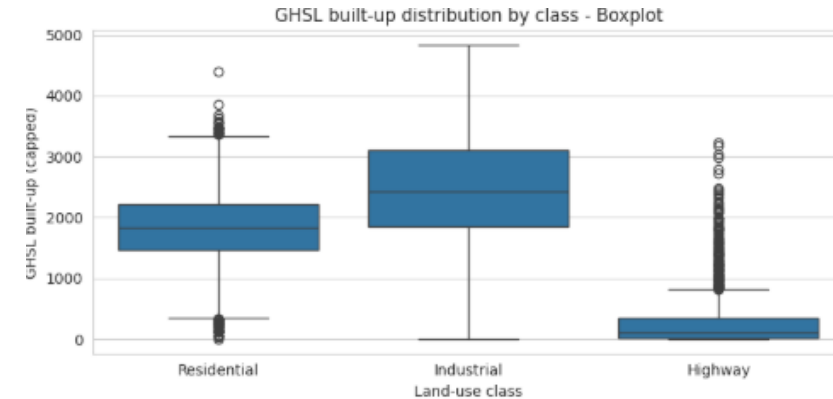
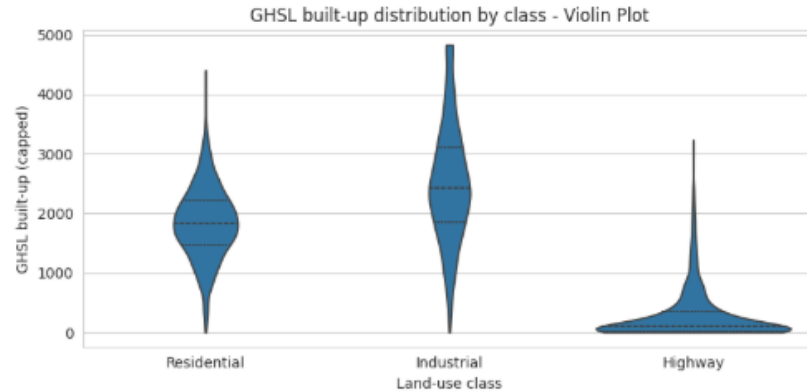
GHSL (Built-up intensity)

- **Global settlement dataset** produced by the **Joint Research Centre (JRC)** of the European Commission
- Provides **built-up surface estimates at ~ 1 km resolution**
- Used as a **continuous proxy of urbanization intensity**

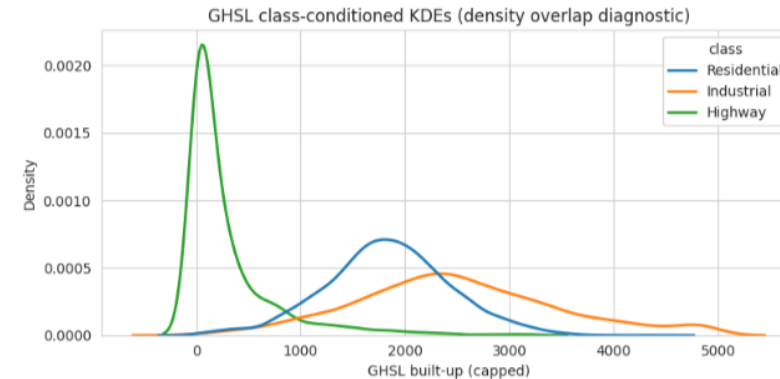
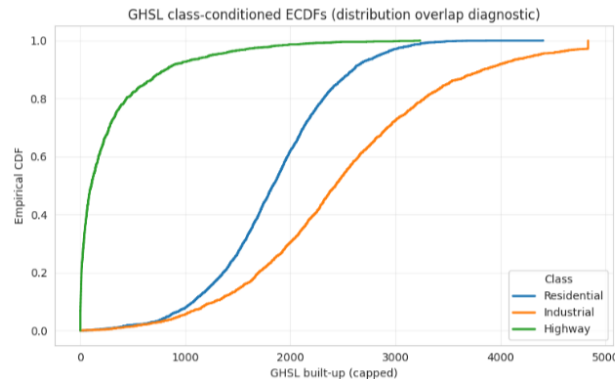


EDA – GHSL by Class

GHSL built-up intensity shows **different central tendencies** across land-use classes (**Industrial > Residential > Highway**). However, **within-class variability** is **high**, with **long tails** in all categories



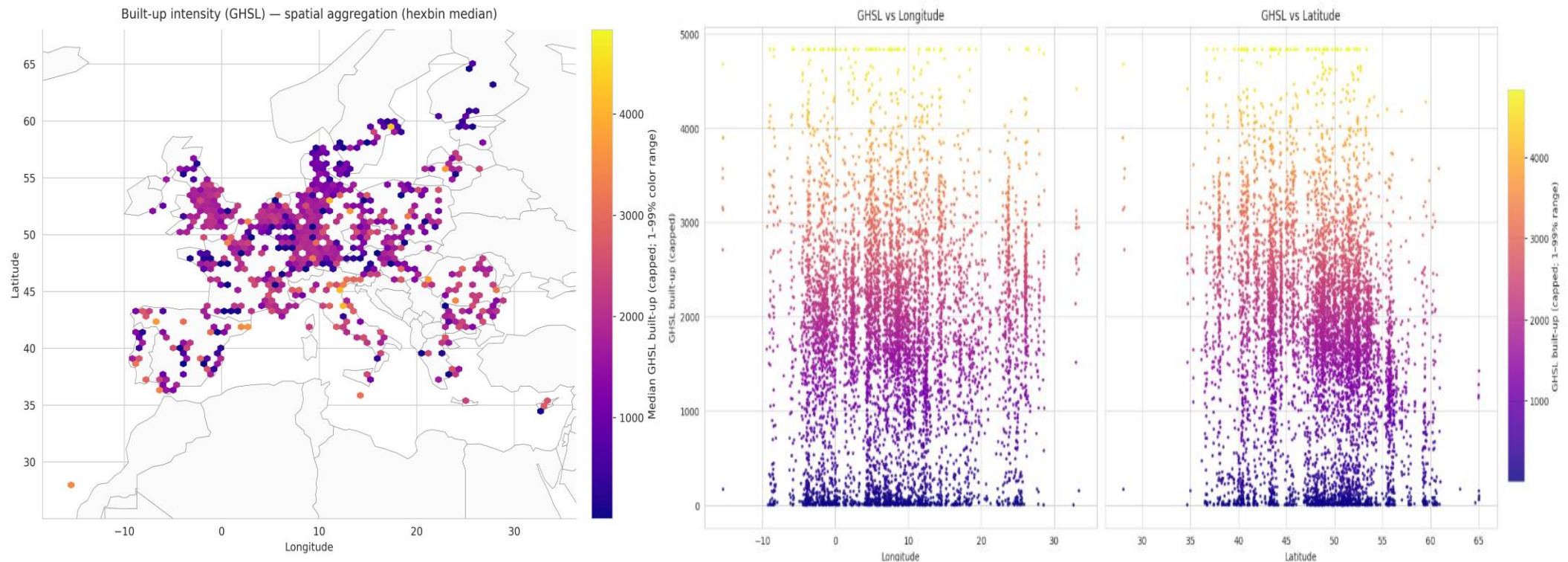
ECDFs and KDEs reveal **strong distributional overlap** between classes, indicating that similar GHSL values occur across different land-use types.



Land-use class provides **partial information**, but GHSL remains **ambiguous** and **noisy** at the patch level.

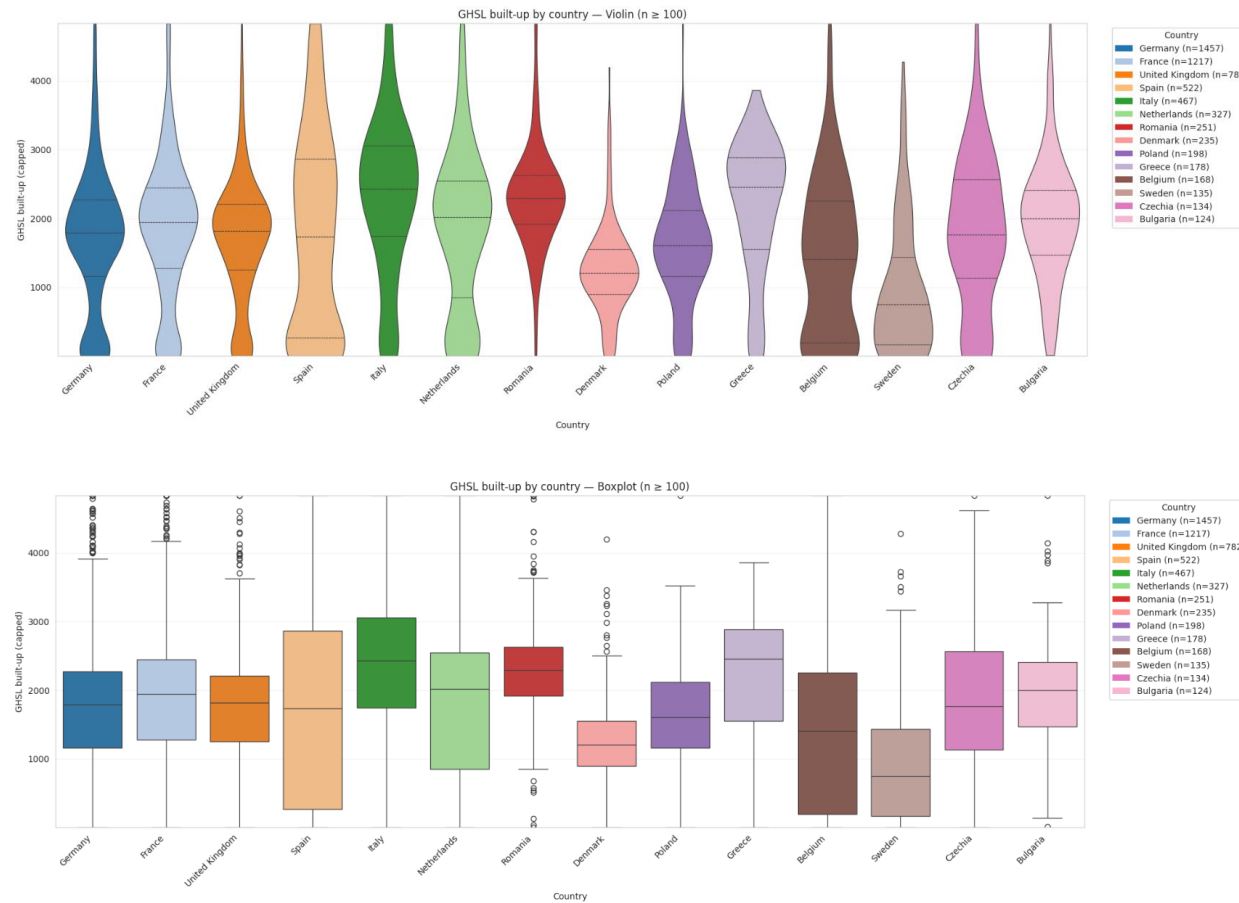
EDA – GHSL Geospatial Analysis

- **High GHSL values** cluster in **major urbanized regions** (Central and Western Europe, Northern Italy, UK)
- **Lower values** dominate **rural and peripheral areas** (Southern and Eastern Europe, Scandinavia)
- **No simple dependence** on **latitude** or **longitude**, since high and low GHSL values appear across the full geographic range



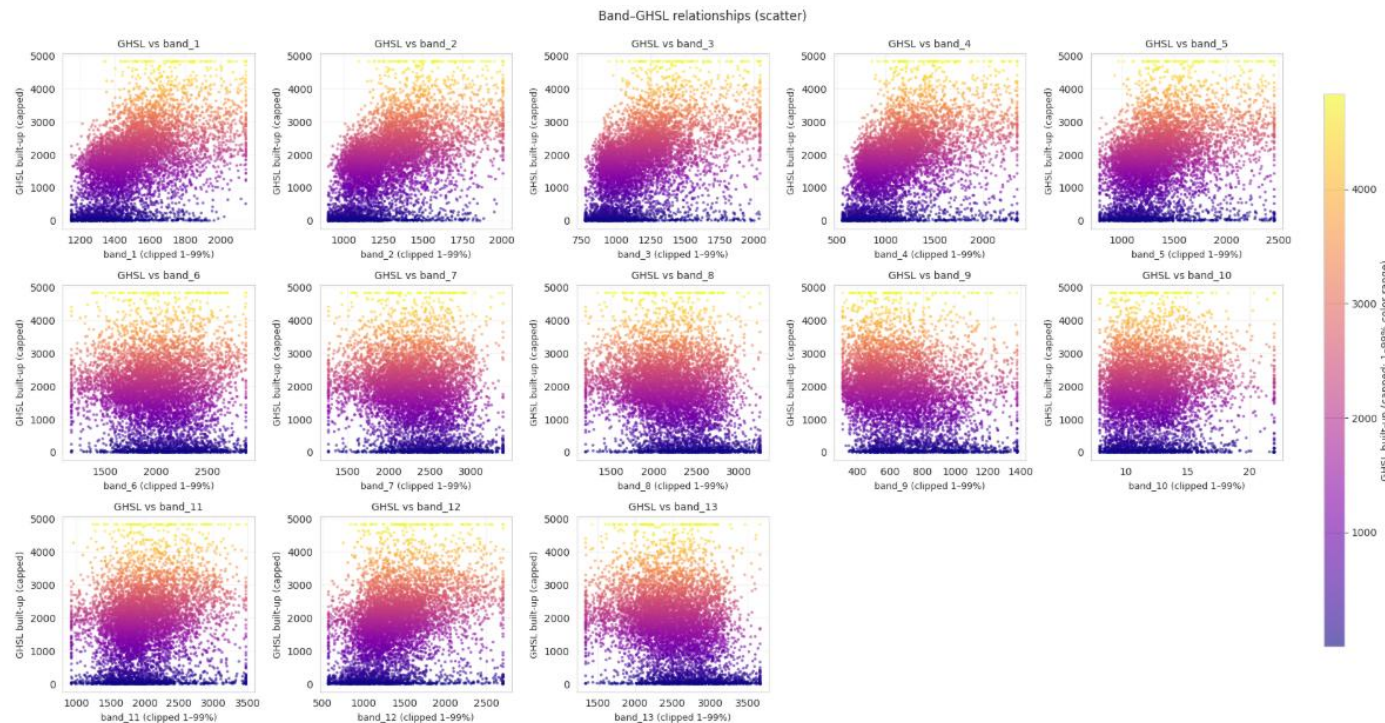
EDA – GHSL Geospatial Analysis

- **Strong local heterogeneity** persists within the same regions
- **Country-level aggregation** captures broad trends, but GHSL distributions remain **wide** and **overlapping** within all countries



EDA – Spectral Predictors

- Each EuroSAT patch is described by **13 spectral bands** measured by the Sentinel-2 sensor (visible, near-infrared and short-wave infrared wavelengths)
- Spectral bands represent **low-level physical measurements**
- **RGB bands (B4, B3, B2)** capture **visible appearance**, while other bands respond to *vegetation, soil, humidity and built structures*



- **Strong overlap** between GHSL values across the full range of each band
- **No** single band provides **clear separability** between *low* and *high* urbanization levels
- Some bands exhibit **weak monotonic trends**, but with *high variability, heteroscedastic behavior, non-linear and overlapping patterns*
- Similar scatter structures across bands indicate **strong inter-band correlations**

The relationship with GHSL is **weak, noisy** and **non-deterministic**, motivating the use of *joint, non-linear* and *representation-based approaches* rather than raw tabular predictors.

EDA – Feature selection

Strategy

Candidate predictors:

Sentinel-2 spectral bands, latitude and longitude

Goal:

Assess how much urbanization can be explained by a compact, non-redundant set of tabular predictors.



Relevance

Spearman correlation

captures *monotonic trends*, robust to skewness and outliers

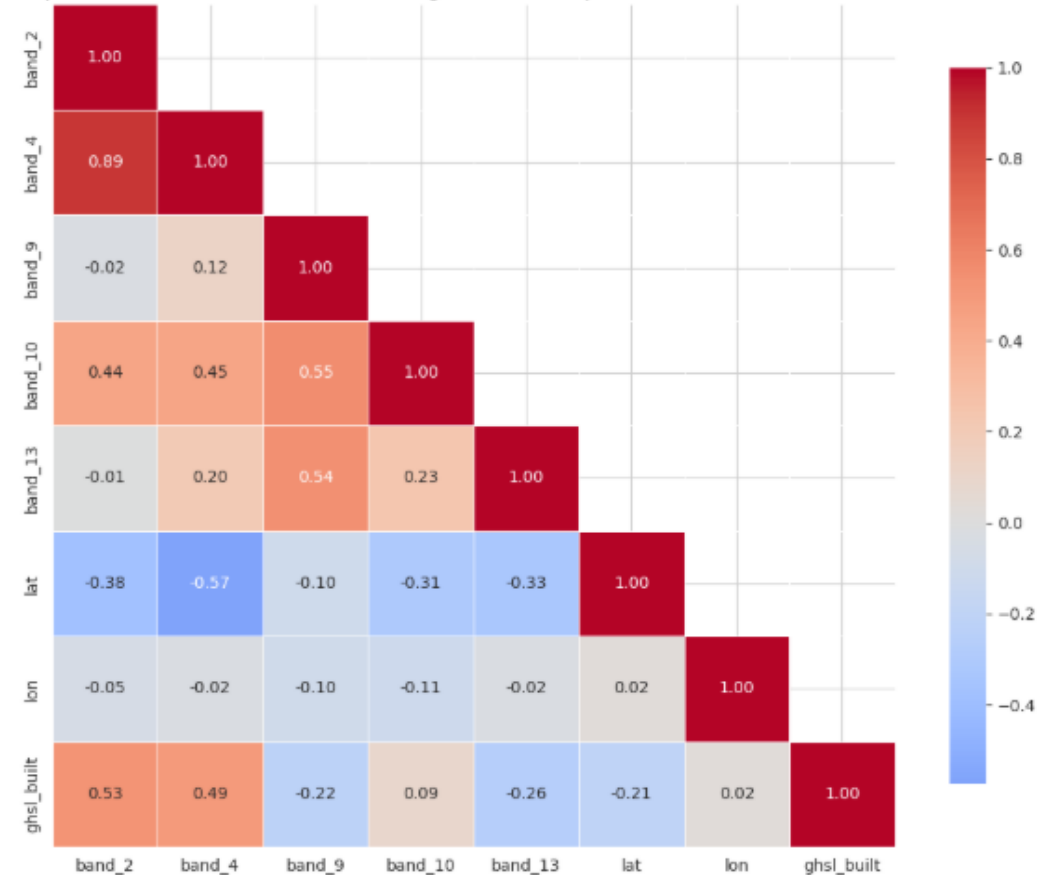
Mutual Information (MI) captures *general non-linear dependencies*



Stability

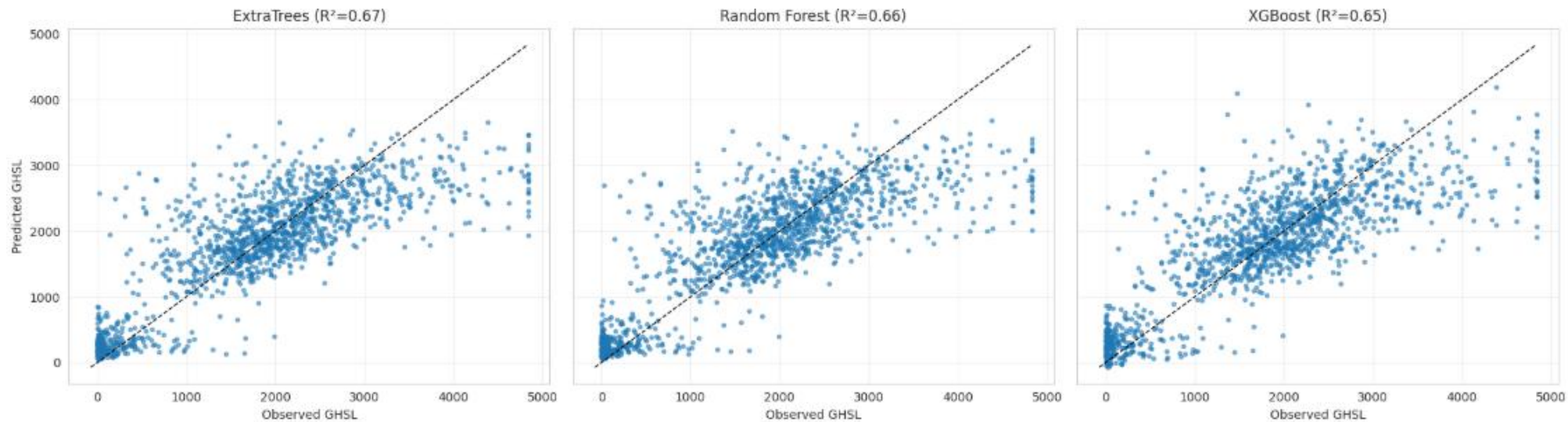
Remove low-variance features and **highly redundant** predictors, keeping the one more related to GHSL

Spearman correlation matrix (lower triangle) — selected predictors, coordinates, and GHSL



EDA – Non-linear regression baselines

- **Models:** Random Forest, ExtraTrees, XGBoost
- **Inputs:** selected spectral bands, latitude, longitude and **coarse land-use semantics** (*dummy variables*)
- **Purpose:** quantify the maximum **explanatory power** of **tabular predictors**



- **Results:** explained variance stabilizes at $R^2 \approx 0.65$ and **large residuals** remain, especially at high GHSL values
- **Key takeaway:** raw spectral and tabular features explain only a **limited fraction of GHSL variability**, indicating that **critical information lies in spatial structure and context**, not in tabular predictors alone.

Model Overview

Motivation

Since Exploratory analysis showed no stable separability of urbanization using tabular proxies alone, we shift to **representation** and **metric learning** directly in the **image domain**.

Goal

Learn an **embedding space** where *visually similar urban morphologies* are **close** and *structurally different areas* are **far apart**.

GHSL-based stratification

GHSL built-up values are split into three strata (**Low** / **Medium** / **High**) using **kNN-informed clustering** ($k = 3$).

Thresholds are defined as midpoints between cluster centroids: $t_1 = \frac{\mu_1 + \mu_2}{2}$, $t_2 = \frac{\mu_2 + \mu_3}{2}$.

This preserves the **ordinal structure of urbanization**.

Model architecture (Siamese network with two identical branches)

ResNet-18 (ImageNet pre-trained) as image encoder,
residual blocks capture multi-scale **spatial patterns**,
global **average pooling**,
final layer projects images into a **128-D embedding**,
embeddings are **L2-normalized** so distances are comparable.

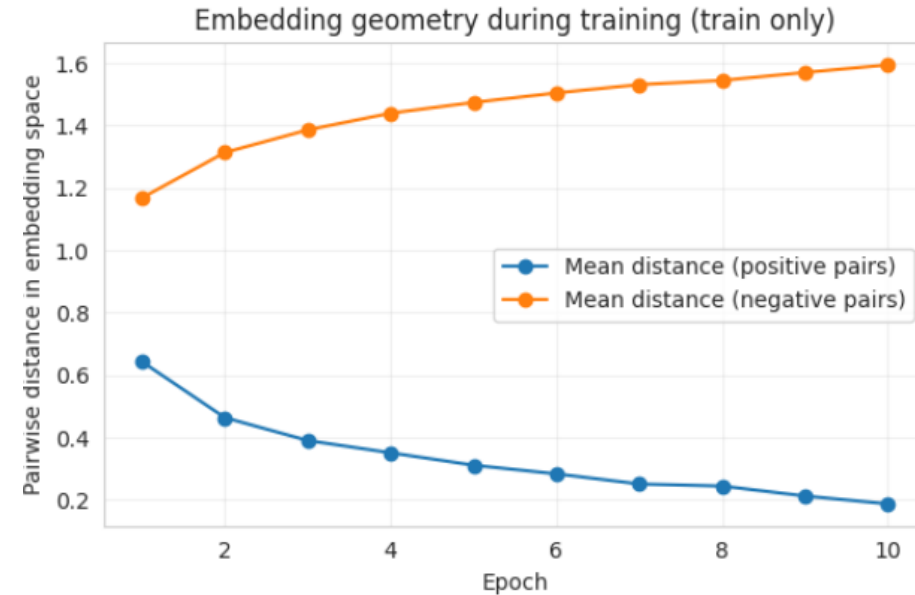
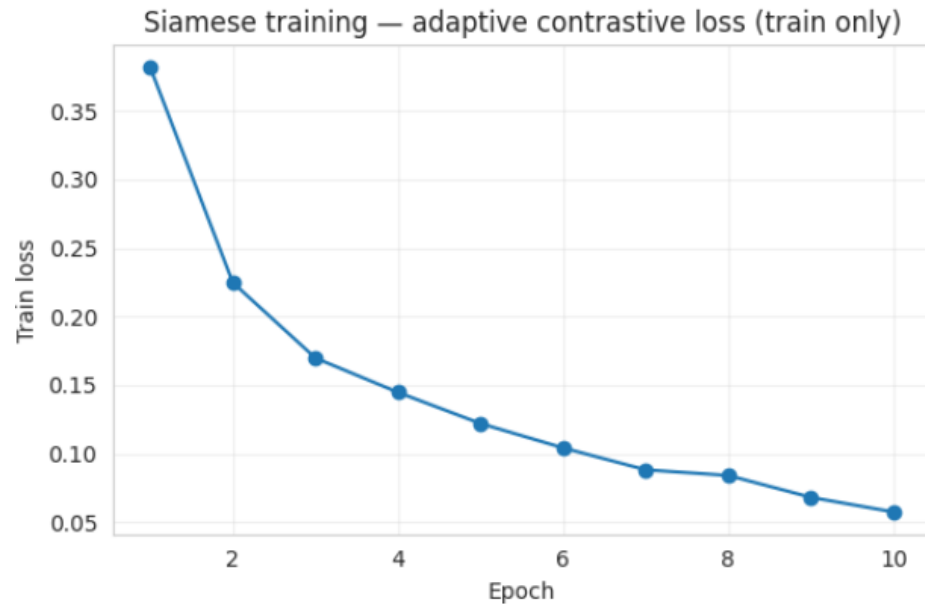


Training mechanism

The model is trained on **pairs of images**. For each pair, images assigned to the **same urbanization stratum** (Low, Medium, High) are treated as *similar* (**positive pairs**) and pulled **closer** in the embedding space, whereas images belonging to **different strata** are treated as *dissimilar* (**negative pairs**) and pushed **apart** by enforcing a *minimum distance* (**margin**).

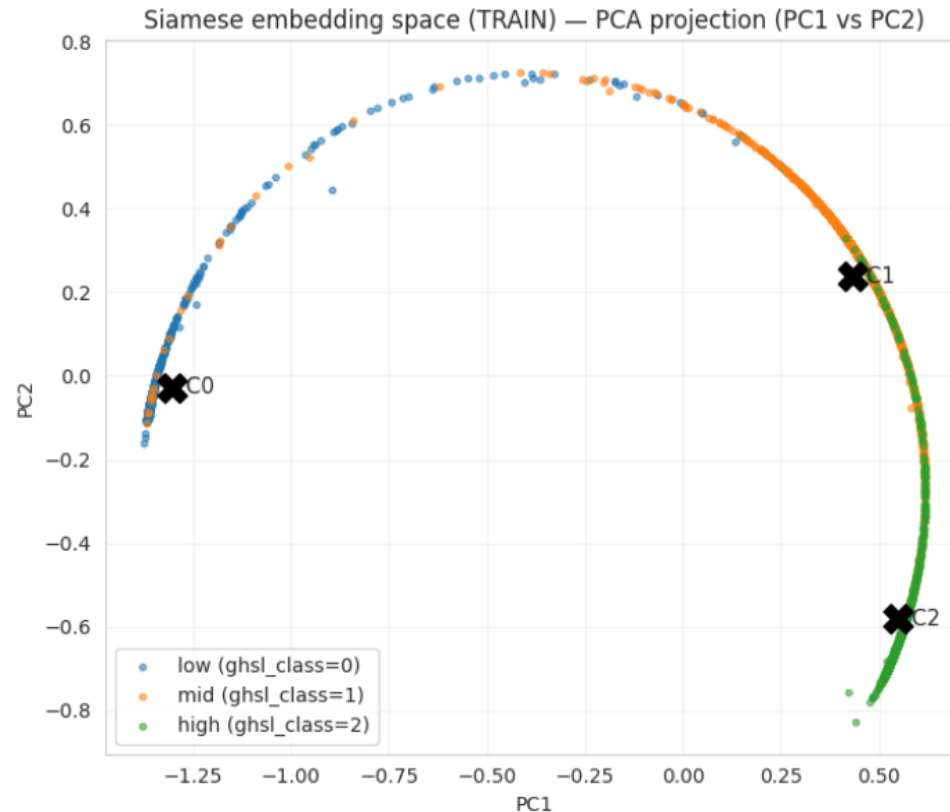
The **margin is adaptive**, being smaller for mildly different urban areas and larger for strongly different ones. It is defined as $m(\Delta g) = \text{clip}(m_0 + m_1 |\log(1 + g_1) - \log(1 + g_2)|)$, where g_1 and g_2 are the *GHSL built-up values* of the two images, m_0 is the *base margin* and m_1 controls the *sensitivity of the margin to differences in urbanization level*.

Adaptive Contrastive Loss: $L = yd^2 + (1 - y)\max(0, m(\Delta g) - d)^2$, where, for a pair of embeddings z_1, z_2 : $d = ||z_1 - z_2||^2$, $y = 1$ if same stratum, 0 otherwise; $m(\Delta g)$ is the **adaptive margin**.



Embedding space analysis

The model produces a **structured latent space of satellite images**, where distances reflect **urban morphology similarity** and embeddings support **ranking by urbanization intensity**.



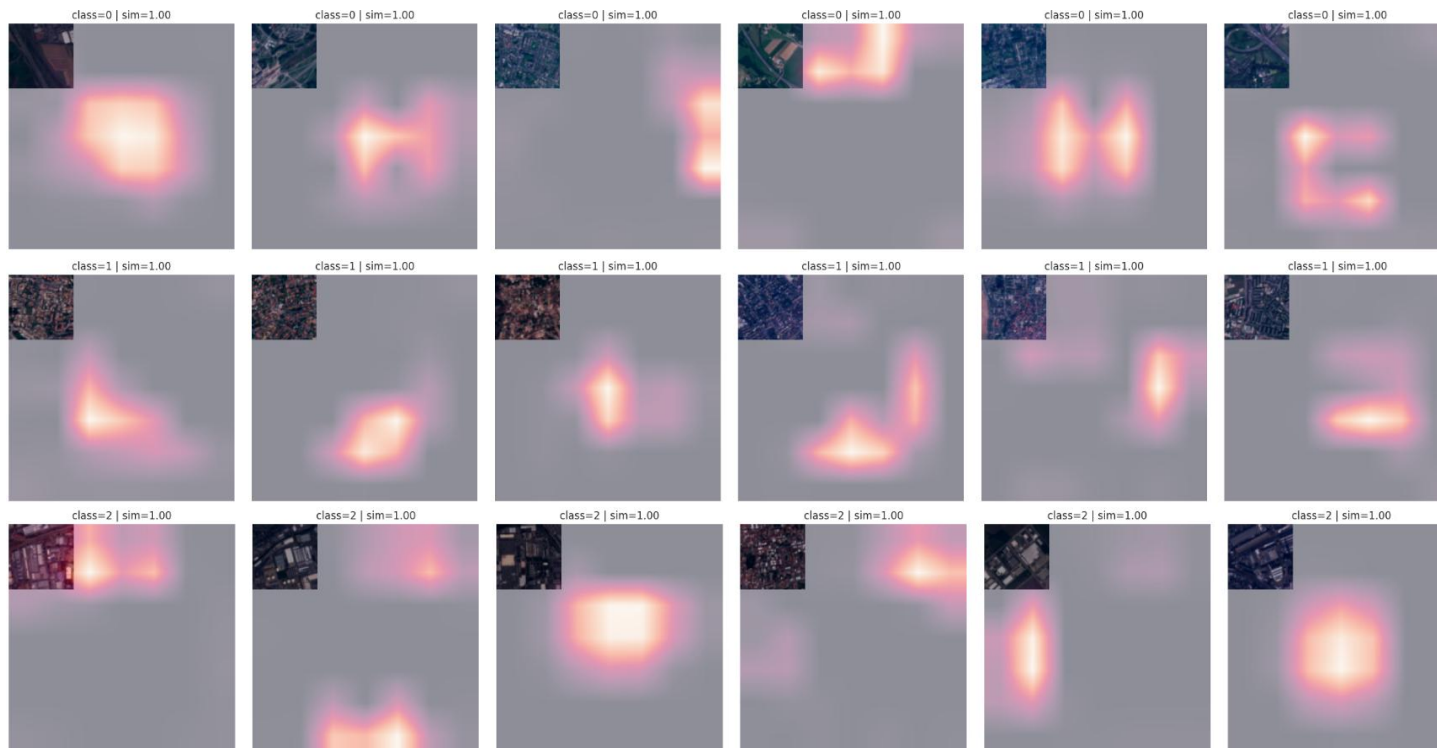
The embedding space reveals **gradual morphological transitions** rather than hard class boundaries, indicating that the model captures an **ordered notion** of **urban morphology**.

This behavior is consistent with the strong global **Spearman correlation** observed between the embedding score and GHSL on both **training** ($\rho \approx 0.88$) and **test** ($\rho \approx 0.79$) sets.

Qualitative interpretation — Grad-CAM

To understand which visual signals drive the learned representation, we compute **Grad-CAM heatmaps** on the Siamese encoder. They reveal **systematic changes in visual attention** across the three GHSL-informed strata.

These patterns indicate that the model learns **morphological and spatial configurations**, consistent with the **strong global Spearman correlation** with GHSL



Diffuse and sparse patterns
(low urbanization)

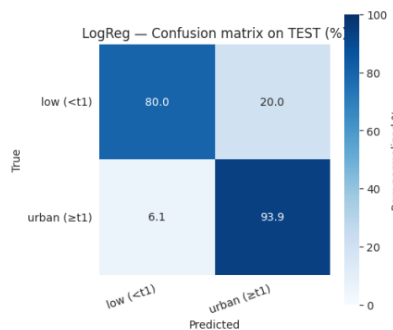
Localized and heterogeneous signals (medium urbanization)

Strong, spatially coherent responses over dense built-up structures (high urbanization)

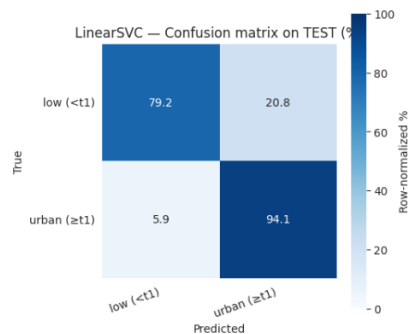
Downstream classification results (Low vs Medium+High)

Multiple standard **classifiers** (LogReg, kNN, SVM, Random Forest, ExtraTrees, XGBoost) were trained **as a diagnostic tool**, to assess whether the *learned embedding space* makes **low vs medium+high urbanization separable**.

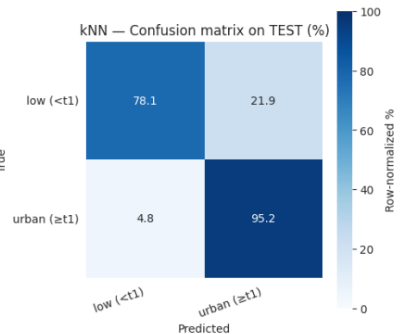
All models achieve very **similar test performance** (≈ 0.90 – 0.91 accuracy, precision ≈ 0.88 – 0.89 , recall ≈ 0.87 , **F1** ≈ 0.87 – 0.88), despite large differences in model complexity. This consistency indicates that **class separability** is **already encoded** in the **embedding space** and performance is driven by the representation rather than by the choice of classifier.



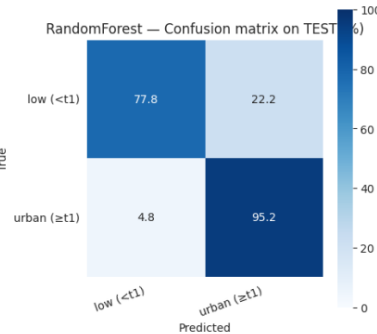
LogReg
Accuracy: 0.902
Precision: 0.878,
Recall: 0.87,
F1: 0.874



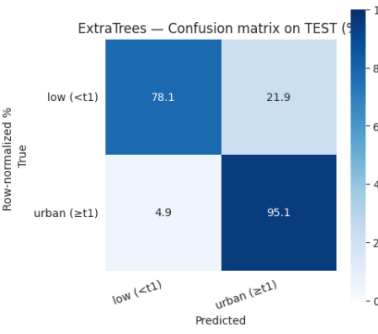
LinearSVC
Accuracy: 0.901,
Precision: 0.878,
Recall: 0.867,
F1: 0.872



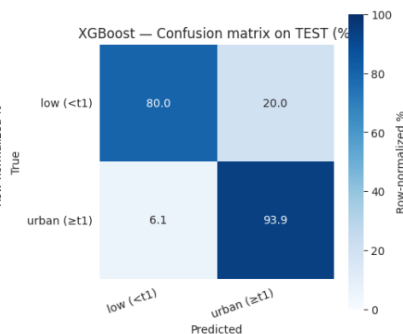
kNN
Accuracy: **0.906**,
Precision: **0.89**,
Recall: **0.867**,
F1: **0.877**



RandomForest
Accuracy: 0.905,
Precision: 0.889,
Recall: 0.865,
F1: 0.876



ExtraTrees
Accuracy: 0.905,
Precision: 0.888,
Recall: 0.866,
F1_macro: 0.876



XGBoost
Accuracy: 0.902,
Precision: 0.878,
Recall: 0.87,
F1_macro: 0.874

Conclusions & Future Work

Conclusions

- Tabular proxies fail to capture urbanization due to missing **spatial and morphological information**.
- Metric learning on satellite images learns an **ordered embedding space** reflecting urban morphology.
- GHSL provides effective **weak supervision**, while downstream results confirm robust separability.
- Grad-CAM shows that the model relies on **meaningful spatial structures**, not trivial cues.

Future Work

- Extend to **ordinal or multi-level prediction** directly in the embedding space.
- Incorporate **additional weak supervision sources**.
- Study **cross-region** and **temporal generalization**.
- Apply embeddings to **urban change** and **risk analysis**.



Thank you for your attention!

